

IET, DAVV, Indore
B.E. I (IT- A & B)
Class Test – II (Nov.' 2023)
1AMRC1: Applied Mathematics-I

Time: 70 min.

Maximum Marks: 20

Note: Attempt only four questions.

- Q:1 If $Y_1 = \frac{X_2 X_3}{X_1}$, $Y_2 = \frac{X_3 X_1}{X_2}$, $Y_3 = \frac{X_2 X_1}{X_3}$ Show that the Jacobian of Y_1, Y_2, Y_3 with respect to X_1, X_2, X_3 is 4. 5M
- Q:2 Expand $f(x, y) = e^x \log(1 + y)$ in power of x & y upto terms of third degree. 5M
- Q:3 Find the length of the arc of the parabola $y^2 = 4ax$ cut off by the line $y = 2x$. 5M
- Q:4 Find the surface area of the cardioid $r = a(1 + \cos \theta)$ revolves about the initial line. 5M
- Q:5 Solve: $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dy dx}{\sqrt{1+x^2+y^2}}$ 5M